

Nithin Govindarajan

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Location: Leuven, Belgium

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Areas of expertise

Numerical linear algebra, Numerical analysis, systems theory, dynamical systems and control, signal processing.

Education

- Oct. 2014 - Dec. 2018 PHD in Mechanical Engineering, University of California in Santa Barbara
Dissertation title: “Periodic approximations and spectral analysis of the Koopman operator: theory and applications”.
Advisors: I. Mezić, S. Chandrasekaran.
- Sep. 2009 - Oct. 2012 MSC in Aerospace Engineering (with distinction), Technische Universiteit Delft
Master thesis: “An Optimal Control Approach for Estimating Aircraft Command Margins”.
Advisors: Q.P. Chu, C.C. de Visser.
- Sep. 2006 - Jul. 2009 BSC in Aerospace Engineering (with distinction), Technische Universiteit Delft

Work experience

- Aug. 2019 - present Postdoctoral researcher, KU Leuven, Belgium
- Nov. 2018 - present Co-founder and technical brain of software startup www.matisse.ai in dentistry
- Feb. 2019 - July 2019 Lecturer Mathematics, University of Amsterdam (UvA), The Netherlands
- Nov. 2012 - May 2013 Junior R&D engineer, National Aerospace Laboratory (NLR), Amsterdam, The Netherlands
- Sep. 2011 - May 2012 Intern, Mission Critical Technologies Inc. (on site at NASA Ames), Moffet Field, CA

Fellowships, honors & awards

- Apr. 2016 CCDC fellowship, Center for Control, Dynamics and Computation, Santa Barbara, CA
- Apr. 2014 Department Merit Fellowship (UCSB Mech. Eng), Santa Barbara, CA
- May 2011 Huygens Scholarship Programme, Nuffic, The Hague, The Netherlands

Publications

Journal Publications

- 2025 Govindarajan, N., Chandrasekaran, S., Dewilde Lathauwer, L. (2025). Tree Quasi-separable Matrices: A Simultaneous Generalization of Sequentially and Hierarchically Semiseparable Representations. *SIAM Journal on Matrix Analysis and Applications*, 46(2), 1562-1586.
- 2024 Govindarajan, N., Widdershoven, R., Chandrasekaran, S., & De Lathauwer, L. (2023). A fast algorithm for computing Macaulay nullspaces of bivariate polynomial systems. *SIAM*

Journal on Matrix Analysis and Applications, 45(1), 368-396.

Govindarajan, N., Vervliet, N., & De Lathauwer, L. (2022). Regression and classification with spline-based separable expansions. *Frontiers in big Data*, 5, 688496.

Govindarajan, N., Epperly, E. N., & Lathauwer, L. D. (2022). $(L_r, L_r, 1)$ -Decompositions, Sparse Component Analysis, and the Blind Separation of Sums of Exponentials. *SIAM Journal on Matrix Analysis and Applications*, 43(2), 912-938.

Epperly, E. N., Govindarajan, N., & Chandrasekaran, S. (2021). Minimal rank completions for overlapping blocks. *Linear Algebra and its Applications*, 627, 185-198.

Govindarajan, N., Mohr, R., Chandrasekaran, S., & Mezic, I. (2021). On the approximation of Koopman spectra of measure-preserving flows. *SIAM Journal on Applied Dynamical Systems*, 20(1), 232-261.

Govindarajan, N., Mohr, R., Chandrasekaran, S., & Mezic, I. (2019). On the approximation of Koopman spectra for measure preserving transformations. *SIAM Journal on Applied Dynamical Systems*, 18(3), 1454-1497.

Govindarajan, N., De Visser, C. C., Van Kampen, E., Krishnakumar, K., Barlow, J., & Stepanyan, V. (2015). Optimal control framework for estimating autopilot safety margins. *Journal of Guidance, Control, and Dynamics*, 38(7), 1197-1207.

Govindarajan, N., de Visser, C. C., & Krishnakumar, K. (2014). A sparse collocation method for solving time-dependent HJB equations using multivariate B-splines. *Automatica*, 50(9), 2234-2244.

Conference proceedings

Widdershoven, R., Govindarajan, N., De Lathauwer, L. (2023, September). Overdetermined systems of polynomial equations: tensor-based solution and application. *Proceedings of EUSIPCO 2023*, Helsinki, Finland.

Chandrasekaran, S., Govindarajan, N., & Rajagopal, A. (2018, July). Fast Algorithms for Displacement and Low-Rank Structured Matrices. In *Proceedings of the 2018 ACM International Symposium on Symbolic and Algebraic Computation* (pp. 17-22).

Govindarajan, N., Arbabi, H., Van Blargian, L., Matchen, T., & Tegling, E. (2016, December). An operator-theoretic viewpoint to non-smooth dynamical systems: Koopman analysis of a hybrid pendulum. In *2016 IEEE 55th Conference on Decision and Control (CDC)* (pp. 6477-6484). IEEE.

Preprints & Tech reports

Widdershoven, R., Govindarajan, N., De Lathauwer, L. (2025). Fast Macaulay null space through the intersection of shifted null spaces. Technical Report 25-56, ESAT-STADIUS, KU Leuven.

Chandrasekaran, S., Epperly, E. N., Govindarajan, N. (2019). Graph-induced rank structures and their representations. *arXiv preprint arXiv:1911.05858*.

Selected talks

“Macaulay matrices, low displacement rank, and the efficient computation of null spaces”, Back to the roots seminar, Leuven.

“A tensor-based approach to solving systems of multivariate polynomials”, CAM23, Selva di Fasano.

“Efficient Computation of Macaulay Matrix Null Spaces Through Exploiting Shift-Invariant Structures”, SIAM AG23, Eindhoven.

“ $(L_r, L_r, 1)$ -decompositions, Sparse Component Analysis, and the Blind Separation of Sums of Exponentials”, SeLMA meeting, Leuven. “Spline-based separable expansions for approximation, regression and classification”, IPAM Workshop I: Tensor Methods and their

2017	Applications in the Physical and Data Sciences, UCLA (online)
2016	“A toolbox for computing spectral properties of dynamical systems”, SIAM DS17, Snowbird. “An operator-theoretic viewpoint to non-smooth dynamical systems: Koopman analysis of a hybrid pendulum”, IEEE CDC 16, Las Vegas.

Teaching

Lead instructor & course organizer

Semester 1 2019/2020	Numerical mathematics, Amsterdam University College
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Co-instructor

March 2022	Fast algorithms for dense structured matrices, KU Leuven
Semester 2 2021/2022	Numerieke modellering & benadering, KU Leuven
Semester 2 2019/2020	Numerieke modellering & benadering, KU Leuven

Teaching assistant

Semester 1 2021/2022	Numerieke wiskunde, KU Leuven
Semester 1 2020/2021	Numerieke wiskunde, KU Leuven
Semester 1 2019/2020	Numerieke wiskunde, KU Leuven
Spring 2018	Control theory, UCSB
Winter 2017	Electrical circuits Lab, UCSB
Fall 2017	Intro to programming in Matlab, UCSB
Summer 2017	Dynamics, UCSB
Summer 2017	Physics Lab: intro to classical mechanics for non-engineers, UCSB
Spring 2017	Dynamics, UCSB
Spring 2015	Dynamics, UCSB
Winter 2015	Vibrations, UCSB
Fall 2014	Statics, UCSB

Extra-curricular

Software skills

Matlab, Python, Julia, Mathematica (basic), C++ (basic), Latex, Git.

Languages

English, Dutch, Tamil (basic).